

Lab Assignment 4b

Hamming-Distance and Similarity Engine

1 Introduction

In Lab Assignment 4a, you designed and verified, in simulation, a combinational circuit that computes the *Hamming distance* between two 8-bit words, the number of bit positions at which they differ, along with the associated equality and threshold-comparison outputs. Recall that for

$$d(10110100, 10011100) = 2.$$

In this assignment you will bring your Lab 4a design onto the Basys3 FPGA. The `hamming_distance8` module itself remains unchanged: it is instantiated as-is inside a new *wrapper* module whose job is to gather the operands from the slider switches, hand them to the distance engine, and present the results on the board’s LEDs and seven-segment displays.

Note: This assignment is hardware-based and will be demonstrated on the Basys3 board.

2 FPGA Demonstration

The Basys3 board provides sixteen slider switches, which is not enough to present both 8-bit operands and the 4-bit threshold simultaneously. The two operands therefore *share* a single bank of eight switches, and a selector switch determines which of the two is currently being written. Use the following mapping for the compulsory demonstration:

Board resource	Function
SW7–SW0	Operand value currently being entered
SW8	Operand select: 0 $\Rightarrow$ write A , 1 $\Rightarrow$ write B
SW12–SW9	Threshold
LED7–LED0	Difference vector $A \oplus B$
LED14	<code>within_threshold</code>
LED15	<code>equal</code>
Seven-segment digits	--0d, where d is the distance

2.1 Capturing the two operands

Since SW7–SW0 carry only one operand at a time, each operand must be stored in a register that retains its value while the other is being entered.

Write a single 8-bit 2:1 multiplexer module and instantiate it as necessary such that SW8 determines which register is written, 0 for register A and 1 for register B.

Do not use an `if/else` construct in the capture logic to route the switch value.

2.2 Driving the display

The four-digit display must show `--0d`, where the two leftmost digits show a dash (segment `G` only), the third digit shows `0`, and `d` is the computed distance. Reuse the seven-segment decoder from Assignment 2 and the anode-multiplexing refresh logic from Assignment 3.

Note that the decoder from Assignment 2 covers the digits `0-9` only; you will need to extend it with the required symbols.

Note: The operator restrictions from Lab 4a apply only to `hamming_distance8` and its submodules, which remain unchanged. The wrapper, capture logic, and display logic may use behavioural constructs freely.

3 Submission and Demo Instructions

Submit the following on Gradescope:

1. all synthesizable Verilog modules, including the board wrapper;
2. a warning-clean constraint file (`.xdc`) and generated bitstream file;
3. a concise report of 2-3 pages

The report must contain a block diagram of the wrapper showing the capture registers, the instantiated `hamming_distance8`, and the display path; a clear identification of which parts of the design are combinational and which are sequential; FPGA resource counts (LUTs, flip-flops, BRAMs, and DSPs); and a brief discussion of the synthesized schematic. Code screenshots without explanation do not constitute a report.

During the assigned lab slot, demonstrate the programmed FPGA and be prepared to explain any module or signal selected by the TA. The submitted source and the demonstrated bitstream must correspond.

4 References

- IEEE document: <https://ieeexplore.ieee.org/document/1620780>
- Basys 3 board reference manual: https://digilent.com/reference/_media/basys3:basys3_rm.pdf